

# Reyyi Shreyas

AI/ML Engineer · LLM Systems, RAG & Evals

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## EDUCATION

### B.E., Artificial Intelligence and Machine Learning

2024 - 2028

BMS Institute of Technology and Management, Bengaluru · CGPA 9.15/10

## EXPERIENCE

### Machine Learning Intern — Launched Global

Aug 2025 - Dec 2025

Delivered SalaryPredict AI end-to-end on the program's ~19.5k-row, 29-feature salary dataset: benchmarked five regression approaches, shipped a stacking ensemble that cut MAE from ₹41,512 (linear regression) to ₹5,791, plus a Flask app with bulk prediction, dashboards and a fairness audit by location, department and education.

1st Place — Launched Global ML Expo (SalaryPredict AI).

## PROJECTS

### Research Paper Analyst — agentic RAG + evaluation harness

[research-agent-rag.streamlit.app](#)

Built an agentic RAG system (LangChain + LangGraph + Gemini) with planner → retriever → verifier stages that routes queries across retrieval, summarisation and comparison tools over five papers; hybrid BM25 + FAISS retrieval with reciprocal rank fusion for grounded, source-cited answers.

Built a RAGAS evaluation harness on a fixed 30-question set measuring faithfulness, context precision and context recall, with dense and hybrid retrieval configs; shipped with a 15-test pytest suite and Dockerfile.

### TRICP — customer retention & churn analytics

[churn-predictor-kappa.vercel.app](#)

Built end-to-end with AI pair-programming (Claude Code) and deployed live on Vercel: FastAPI backend with interactive OpenAPI docs + React/Vite frontend, retention-email automation, bulk campaigns and intervention simulation over a stacking ensemble (7,043 customers; ROC AUC 0.827).

### ChessMind AI — adaptive chess coach with self-hosted LLM

[github.com/reyyishreyas/ChessMind\\_AI](https://github.com/reyyishreyas/ChessMind_AI)

8-model stacking ensemble on 50,028 move samples predicting per-move Elo change with a game-level split (no game across train/test): held-out  $R^2$  0.959, MAE 0.76 Elo.

Serves ~160 ms/move via FastAPI; cut first-feedback latency 7.6 s → 1.7 s (streaming, caching); CI runs lint, typecheck, 63 tests and build on every push; Stockfish-validated suggestions explained by a local Ollama LLM.

## RESEARCH

First author, "GCM-HAIRNet: A Geographic Context Multimodal Deep Learning Framework for Geospatial Hazard Risk Prediction" — abstract accepted at SICE 2026; full manuscript under review.

Designed the transformer-based Geographic Context Module (SwinV2 + CNN fusion) over a curated 60+ city dataset of GIS and visual features;  $R^2$  0.954 on held-out cities vs 0.627 image-only, outperforming 7

fusion and attention baselines.

## LEADERSHIP

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### ASTRA (defence-tech student club), BMSIT&M — President

Sep 2026 - Present

Technical Head · Sep 2025 – Sep 2026

Led a 30-member team through 6 research manuscript submissions (3 accepted at SICE 2026, incl. my first-author work); ran Git and AI tooling sessions.

## OPEN SOURCE

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Three merged pull requests across established scientific Python projects: decoder training-log fix (nilearn #6614), HBR tutorial docs (PCNtoolkit #551), I/O-guide docs (movement #1115).

## SKILLS

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**Agentic & LLM:** LangChain, LangGraph, MCP (FastMCP), RAG, FAISS, BM25, embeddings, RAGAS, Gemini API, Ollama, tool calling

**ML & Deep Learning:** PyTorch, Scikit-learn, XGBoost, LightGBM, ensemble methods, computer vision

**Backend & Data:** Python, FastAPI, Flask, REST APIs, Docker, SQL, Pandas/NumPy, Git/GitHub, CI/CD

**Tooling:** Claude Code (AI-assisted development), Vercel, Streamlit, pytest